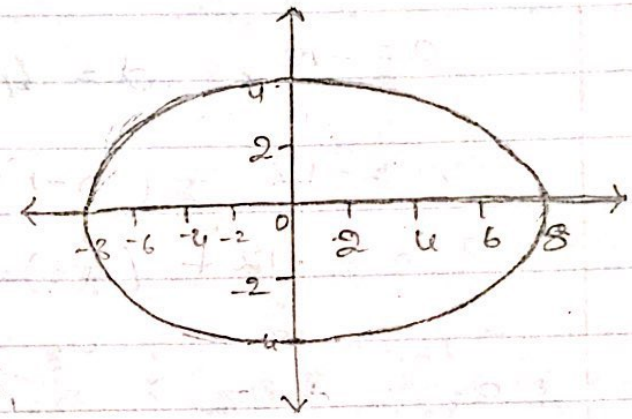
Homework #6

MATH 372

(1)



The endpoint of the major axis are $(-8/2, 0)$ and $(\frac{8}{2}, 0)$
$= (\pm 4, 0)$

The end point of the minor axis are $(0, -4/2)$ and $(0, 4/2) = (0, \pm 2)$

$\therefore a^2 = 4 \times 4 = 16$, $b^2 = 2 \times 2 = 4$.

$c^2 = a^2 - b^2 \Rightarrow c^2 = 16 - 4 \Rightarrow c^2 = 12 \Rightarrow \boxed{c = \sqrt{12}}$

$c \approx 3.4641 \approx$ 3 feet 5 and $9/16$ inches

The string length $= 2a \Rightarrow 2a = 2 \times 4 \Rightarrow$ 8 feet

From the center point of the board each tack 3 feet 5 and $9/16$ inches towards the short sides of the board . The length of the string is 8 feet.

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② $9x^2 - 4y^2 = 36$

$\div 36 \Rightarrow \quad \frac{9x^2}{36} - \frac{4y^2}{36} = 1$

$\Rightarrow \quad \frac{x^2}{4} - \frac{y^2}{9} = 1$ ∴ Equation of Hyperbola.

$a^2 = 4 \Rightarrow \boxed{a = 2}$, $b^2 = 9 \Rightarrow \boxed{b = 3}$

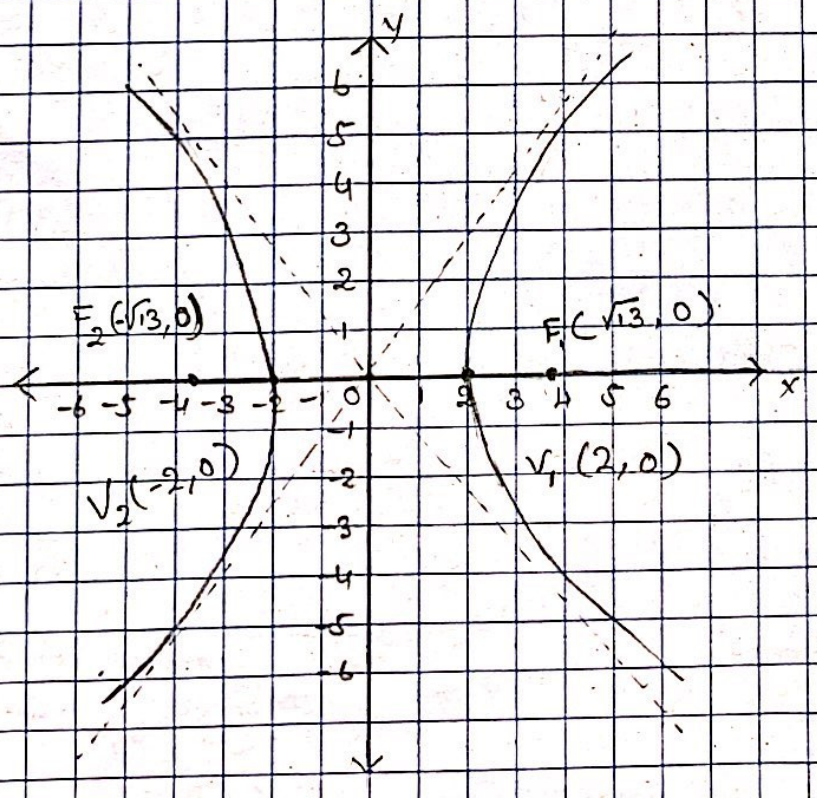
$c^2 = a^2 + b^2 \Rightarrow c^2 = 4 + 9 \rightarrow c^2 = 13 \Rightarrow \boxed{c = \sqrt{13}}$

∴ $\boxed{\text{Center} : (0,0)}$

$\boxed{\text{Vertices} : (\pm 2, 0)} \Rightarrow (-2,0)$ and $(2,0)$

$\boxed{\text{Foci} : (\pm \sqrt{13}, 0)} \Rightarrow (-\sqrt{13}, 0)$ and $(\sqrt{13}, 0)$

Asymptotes : $y = \pm \frac{b}{a}x \Rightarrow \boxed{y = \pm \frac{3x}{2}}$



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(3) $4x^2 - 4x - 8y + 9 = 0$

$4x^2 - 4x = 8y - 9$

$4(x^2 - x) = 8y - 9$

$4\left(x^2 - x + \frac{1}{4}\right) = 8y - 9 + 1$

$4\left(x - \frac{1}{2}\right)^2 = 8(y - 1)$

$\left(x - \frac{1}{2}\right)^2 = 2(y - 1)$

$4P = 2, \Rightarrow \boxed{P = \frac{1}{2}}$

vertex : $\left(\frac{1}{2}, 1\right)$

Since only one variable is squared, the equation $4x^2 - 4x - 8y + 9 = 0$ is parabola.

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④

$$13x^2 + 6\sqrt{3}xy + 7y^2 = 16$$

$$\cot 2\theta = \frac{A - C}{B}$$

$A = 13$, $B = 6\sqrt{3}$, $C = 7$

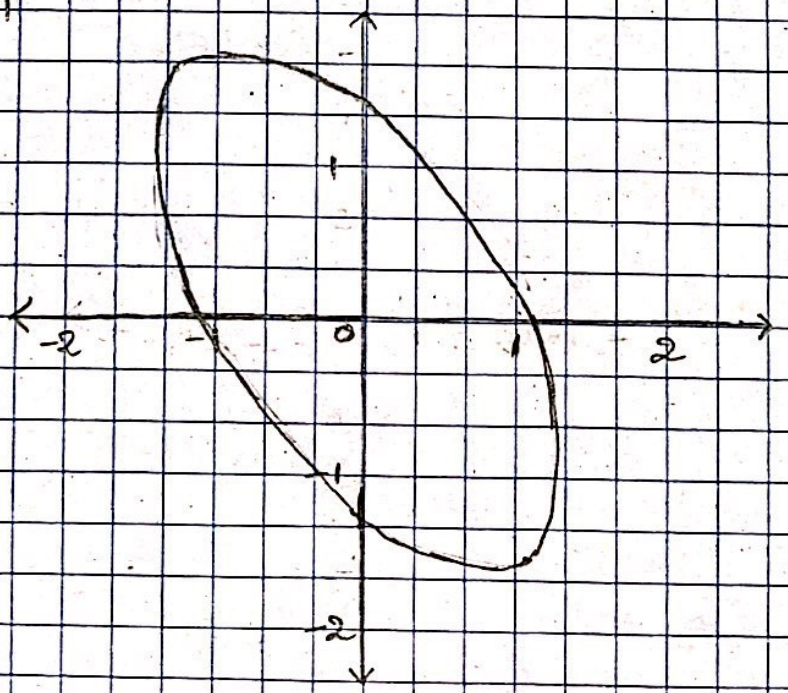
$$\cot 2\theta = \frac{13 - 7}{6\sqrt{3}} \Rightarrow \cot 2\theta = \frac{1}{\sqrt{3}}$$

$$\Rightarrow 2\theta = \frac{\pi}{3} \Rightarrow \boxed{\theta = \frac{\pi}{6}}$$

$$x = X\cos\theta - Y\sin\theta \Rightarrow \boxed{x = \frac{\sqrt{3}}{2}X - \frac{1}{2}Y}$$

$$y = X\sin\theta + Y\cos\theta \Rightarrow \boxed{y = \frac{1}{2}X - \frac{\sqrt{3}}{2}Y}$$

$$\Rightarrow \frac{X^2}{3} + \frac{Y^2}{7} = 1$$



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(5) $a_n = \frac{a_{n-1}}{2}$ and $a_1 = -8$

$n=2 \Rightarrow a_2 = \frac{a_{2-1}}{2} \Rightarrow a_2 = \frac{a_1}{2} \Rightarrow a_2 = -\frac{8}{2}$

$\Rightarrow \boxed{a_2 = -4}$

$n=3 \Rightarrow a_3 = \frac{a_{3-1}}{2} \Rightarrow a_3 = \frac{a_2}{2} \Rightarrow a_3 = \frac{-4}{2}$

$\Rightarrow \boxed{a_3 = -2}$

$n=4 \Rightarrow a_4 = \frac{a_{4-1}}{2} \Rightarrow a_4 = \frac{a_3}{2} \Rightarrow a_4 = \frac{-2}{2}$

$\Rightarrow \boxed{a_4 = -1}$

$n=5 \Rightarrow a_5 = \frac{a_{5-1}}{2} \Rightarrow a_5 = \frac{a_4}{2} \Rightarrow \boxed{a_5 = -\frac{1}{2}}$

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(6)

The arithmetic sequence 11, 8, 5, 2

$$a_n = a + (n-1)d$$

$\boxed{a = 11}$, $\boxed{d = -3}$

$$\boxed{a_n = 11 + (-3)(n-1)}$$

$\boxed{n=5} \Rightarrow a_5 = 11 + (-3)(5-1) \Rightarrow a_5 = 11 - 12 \Rightarrow \boxed{a_5 = -1}$

$\boxed{n = 100} \Rightarrow a_{100} = 11 + (-3)(100-1) \Rightarrow a_{100} = 11 - 297$

$\Rightarrow \boxed{a_{100} = -286}$

The fifth term : $a_5 = -1$

The 100th term : $a_{100} = -286$

The common differences :- $a_n = 11 + (-3)(n-1)$

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